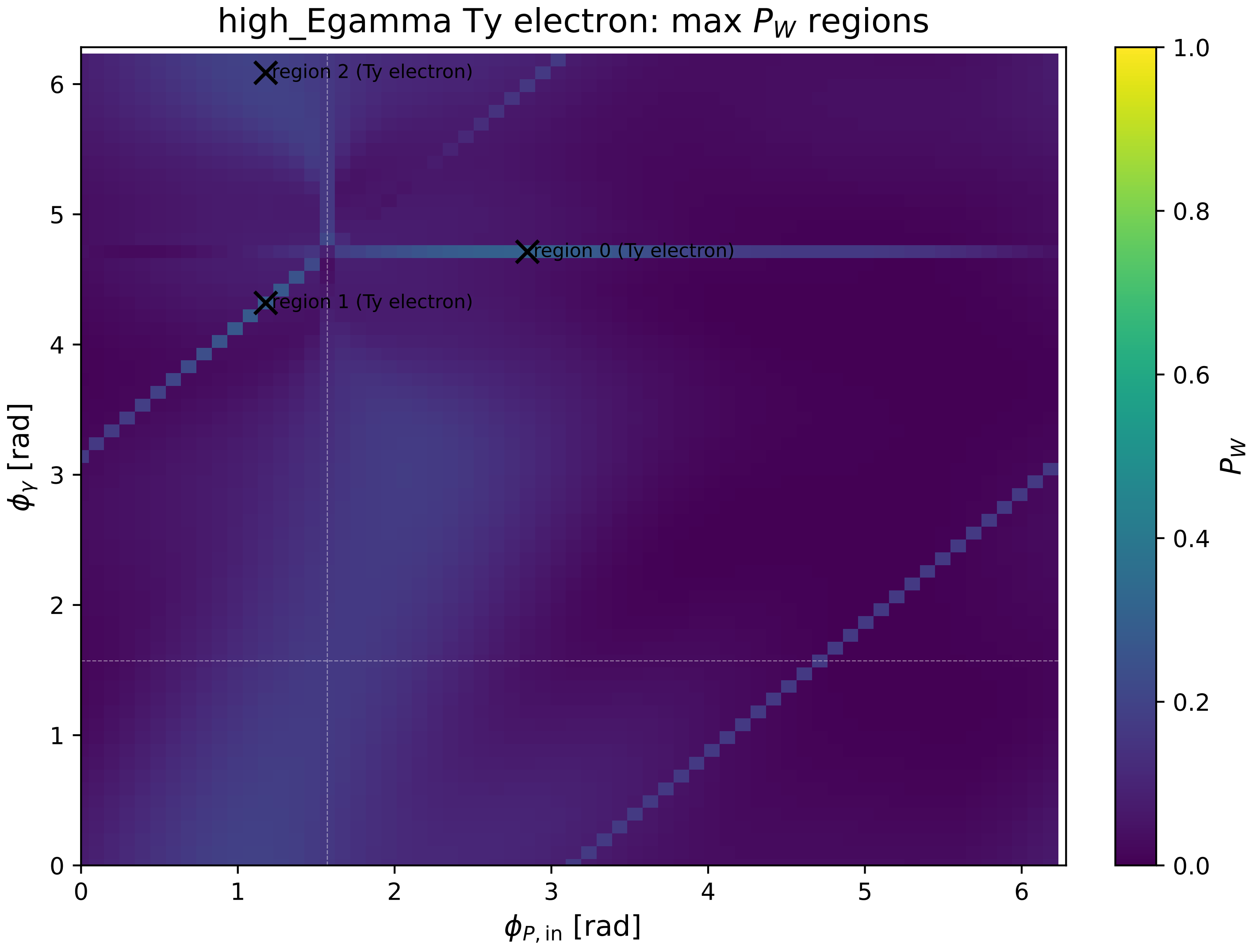
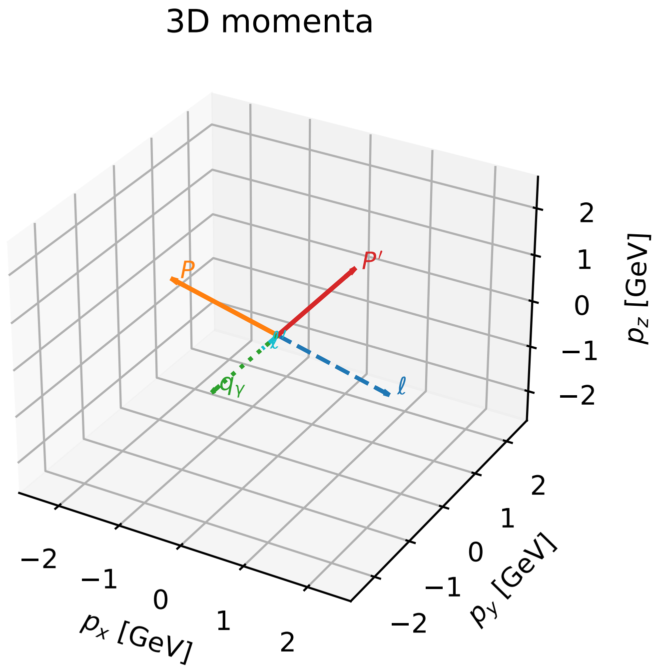


high_Egamma Ty electron: max P_W regions
region 2 (Ty electron)
region 0 (Ty electron)
region 1 (Ty electron)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
P_W
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

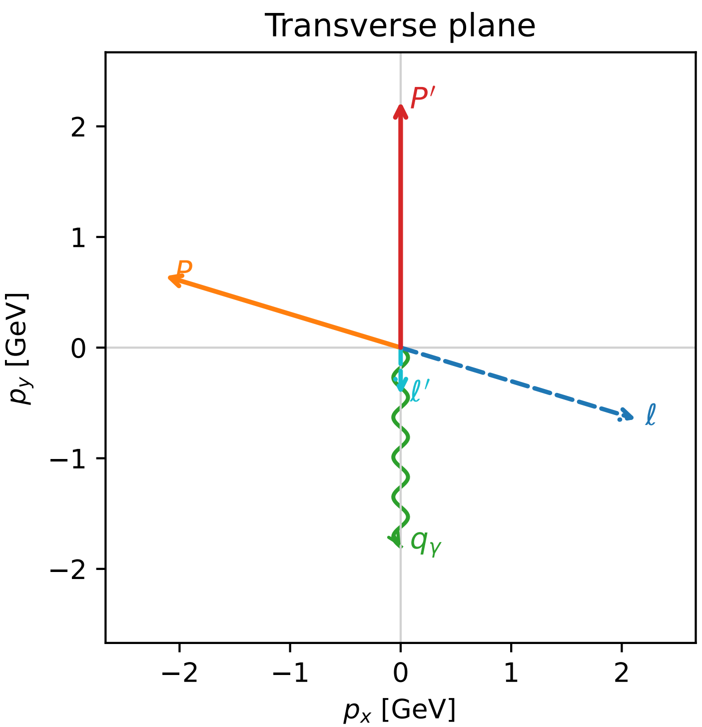
Ty electron characteristic max P_W configuration

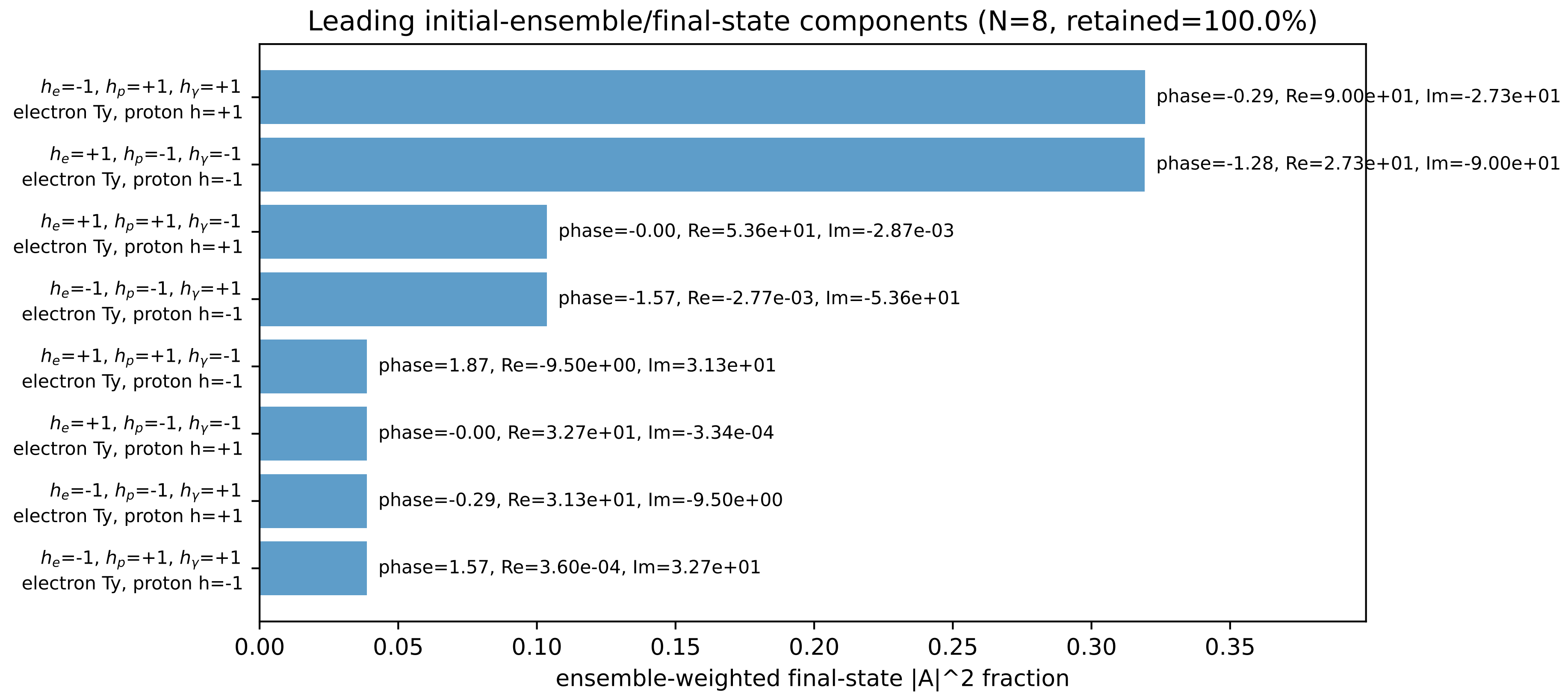


w_purity_high_egamma_ty_lepton_region_0 P_W=0.307237
region: high_Egamma_Ty_lepton_region_0
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,e},h_p)$

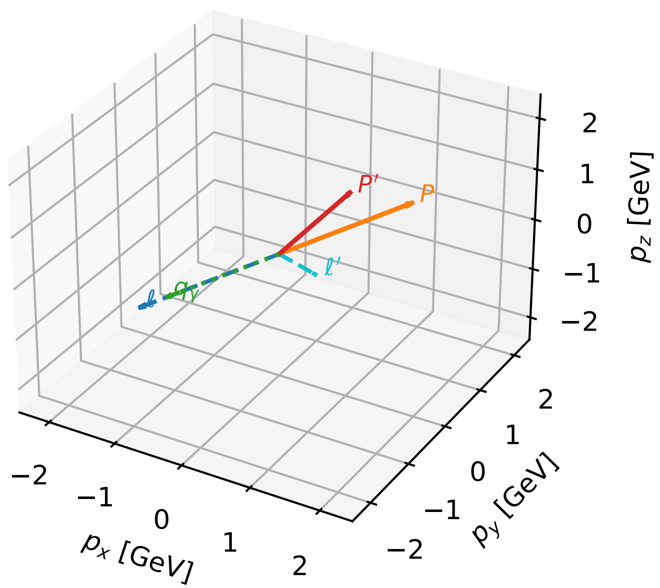
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.22442
$\theta_{\rm in}$=1.57079, ϕ_ℓ=5.98866, ϕ_P=2.84707
E_γ=1.8, ϕ_γ=4.71239
Q^2=1.34006, x_B=0.0742879, t=-7.02339, W^2=17.5785, y=0.874138
$\theta(\ell',\gamma)$=0 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.1286 p_y= -0.6457 p_z= -0.0000
P E= 2.4141 p_x= -2.1286 p_y= 0.6457 p_z= 0.0000
ℓ' E= 0.4244 p_x= 0.0000 p_y= -0.4244 p_z= 0.0000
P' E= 2.4141 p_x= 0.0000 p_y= 2.2244 p_z= 0.0000
q_γ E= 1.8000 p_x= -0.0000 p_y= -1.8000 p_z= 0.0000





Ty electron characteristic max P_W configuration

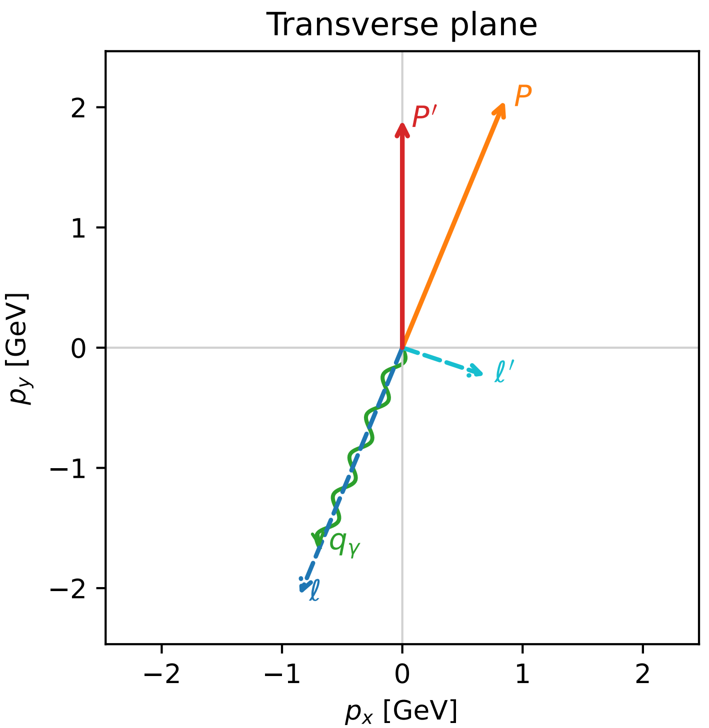


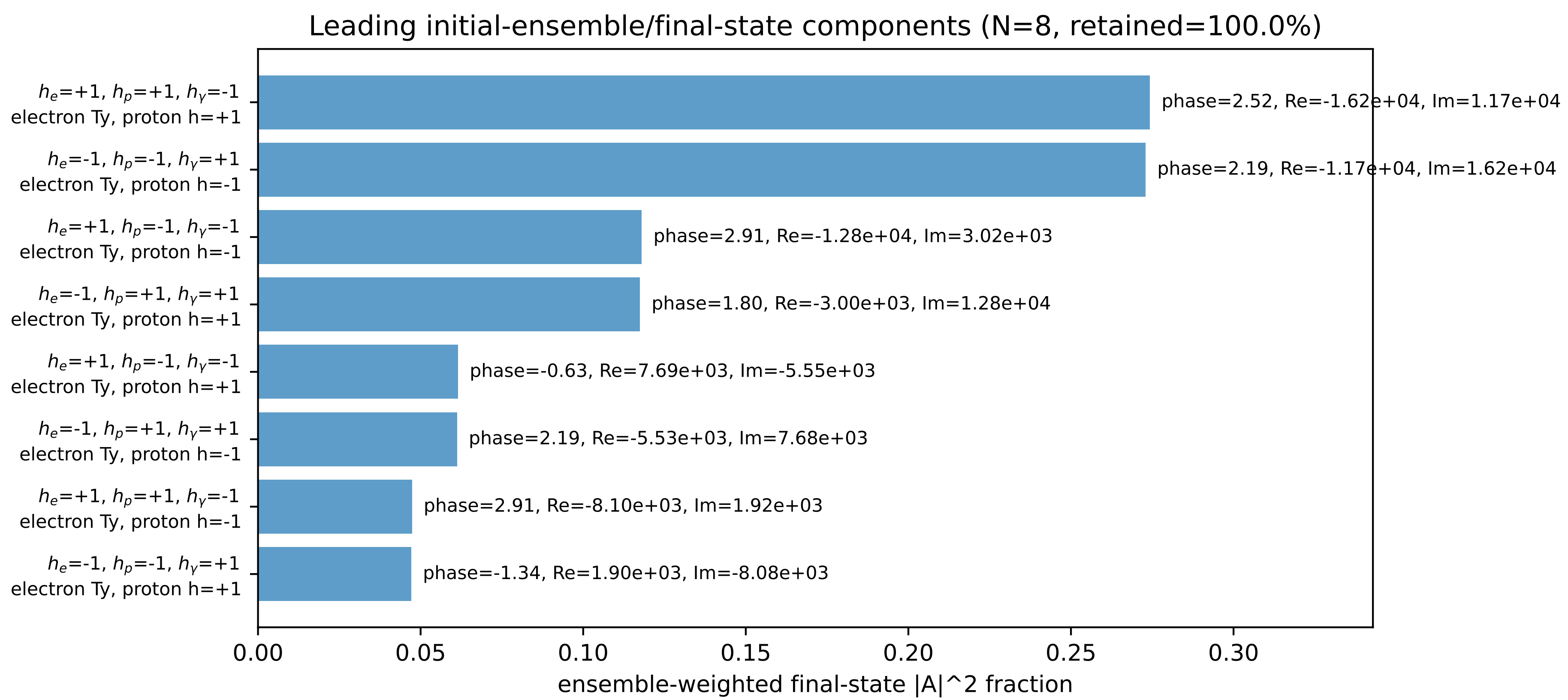

```
w_purity_high_egamma_ty_lepton_region_1  P_W=0.283798
region: high_Egamma_Ty_lepton_region_1
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_high_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_p} \rho(T_{y,e}, h_p)$

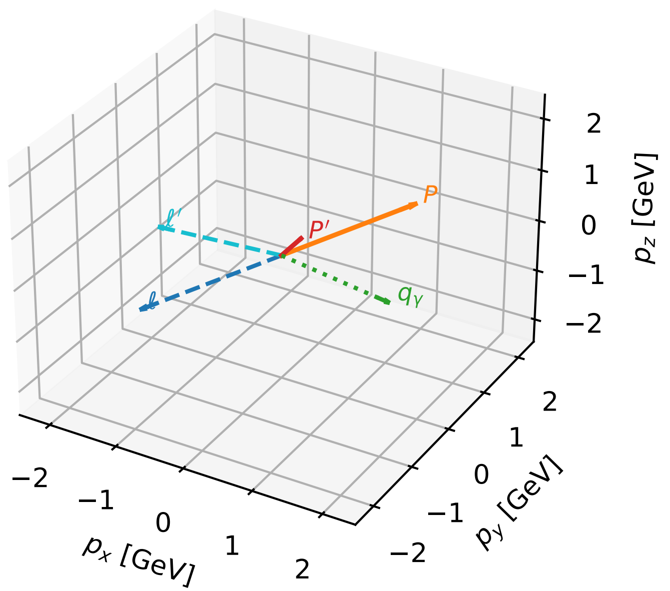
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.89271
$\theta_{\rm in}$=1.57079, ϕ_ℓ=4.31969, ϕ_P=1.1781
E_γ=1.8, ϕ_γ=4.31969
Q^2=3.45894, x_B=0.199263, t=-0.659963, W^2=14.7795, y=0.841182
$\theta(\ell', \gamma)$=94.0564 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -0.8512 p_y= -2.0551 p_z= -0.0000
P E= 2.4141 p_x= 0.8512 p_y= 2.0551 p_z= 0.0000
ℓ' E= 0.7261 p_x= 0.6888 p_y= -0.2297 p_z= 0.0000
P' E= 2.1124 p_x= 0.0000 p_y= 1.8927 p_z= 0.0000
q_γ E= 1.8000 p_x= -0.6888 p_y= -1.6630 p_z= 0.0000





Ty electron characteristic max P_W configuration



w_purity_high_egamma_ty_lepton_region_2 P_W=0.195737
region: high_Egamma_Ty_lepton_region_2
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,e}, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.506977
θ_{in}=1.57079, ϕ_ℓ=4.31969, ϕ_P=1.1781
E_γ=1.8, ϕ_γ=6.08684
Q^2=4.23853, x_B=0.502609, t=-1.30456, W^2=5.07438, y=0.408658
$\theta(\ell', \gamma)$=163.706 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -0.8512 p_y= -2.0551 p_z= -0.0000
P E= 2.4141 p_x= 0.8512 p_y= 2.0551 p_z= 0.0000
ℓ' E= 1.7723 p_x= -1.7654 p_y= -0.1558 p_z= 0.0000
P' E= 1.0662 p_x= 0.0000 p_y= 0.5070 p_z= 0.0000
q_γ E= 1.8000 p_x= 1.7654 p_y= -0.3512 p_z= 0.0000